Chapter → 1

Circuit Analysis

* Circuit → Web of element which forms a closed loop.

* Circuit element → Resistor, Inductor and capacitor

→ Resistor → $V = IR$ (By ohm's Law)
(resistance) ← property of resistor

→ Inductor →
symbol → "~~~~"

$\Psi = N\phi$
$\Psi \propto I$
$N\phi \propto I$
$N\phi = LI$

$L = \frac{N\phi}{I}$

→ Capacitor →
Symbol - "⊣⊢"

$Q \propto V$
$[Q = CV]$
↓ → capacitance
charge

Network → Web of elements which may or may not form closed loop are called Network.

Input excitation → [Circuit or Network] → output Response

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(#) Basic Quantities :→

a) Current (I) or (i) →

$$I = \frac{dq}{dt}$$ → unit - amperes (A)

Rate of flow of charge

b) Voltage → Energy per unit charge between two points, in volts.

$$V = \frac{Kq}{r} \quad \text{or} \quad V = \frac{dw}{dq}$$

c) Power → Energy per unit time

$$P = \frac{dw}{dt} \qquad P = \frac{dw}{dt} \times \frac{dq}{dq} = P = v.i$$

or

$$P = \frac{v^2}{r}$$

unit :- watts (w)

d) Energy → $w = \int P \cdot dt$

(#) Types of elements →

a) Active elements :-

The element which generate or modify the signals are known as active elements.

ex→ Diode, Transistor, Amplifier, Battery etc.

b) Passive elements →

Those elements which are unable to change (generate) or modify the signal are called passive elements.

ex→ Inductor, Transmitor, Capacitor etc.

(#) Linear and Non-Linear elements →

a) Linear → The elements that characteristics are straight line and passing through origin is known as linear elements.

ex→ Resistor, Inductor, Capacitor etc.

a) Biletral

b) ~~Non-Linear~~ → Those elements that offers different in in both direction of current flow.

ex→ Diode, HCR etc.

b) Non-Linear → Those element that characteristics not a straight line is know as non-linear elements.

ex→ Diode, HCR etc.

HCR → High Rupturing Capacity fuse

(#) Unilitral and Biletral elements:→

b) Biletral → The elements that offers some in both the direction of current flow.

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SOURCE

Depend on

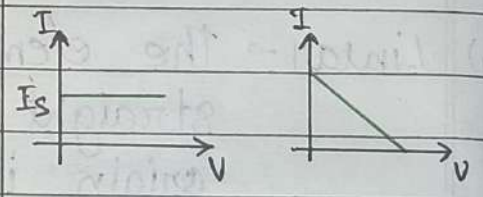
(Current) Dependent Source

- voltage dependent voltage source — $K_x V_x$
- current dependent voltage source — $K_y I_y$
- current dependent current source — $K_y I_y$
- voltage dependent current source — $K_x V_x$

(Voltage) Independent Source

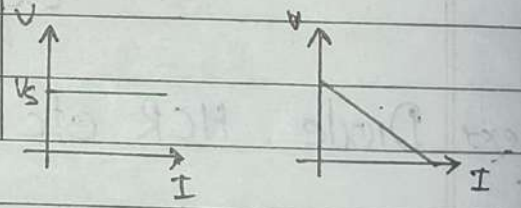
→ Current Source

- Ideal — $R_{SH} = \infty$
- practical — R_{SH}



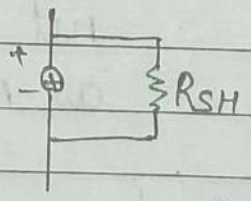
→ Voltage Source

- Ideal — $R_S = 0$
- Practical — R_S



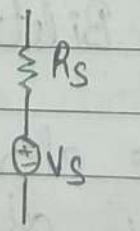
(#) **Current Source** →

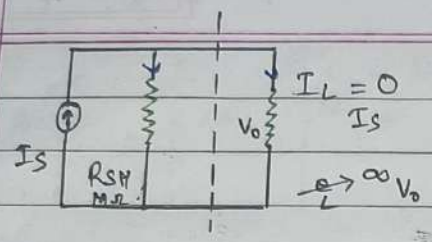
$$\left[R_{SH} = \frac{V_{oc}}{I_{sc}} \right]$$

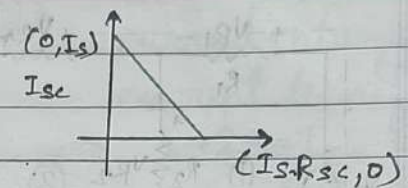


(#) **Voltage Source** →

$$\left[R_S = \frac{V_{oc}}{I_{sc}} \right]$$







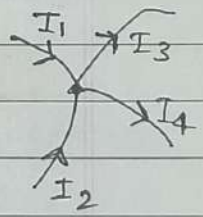
$I_L = I_S = I_{sc}$

$R_{sc} = \frac{\text{O.C. Voltage}}{\text{S.C. current}}$

Kirchoff's Current Law (KCL) →

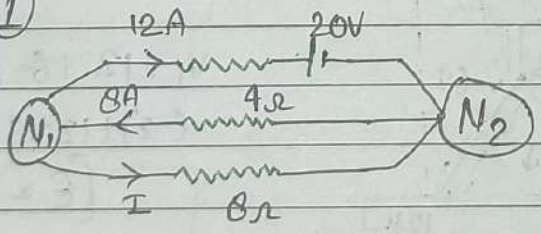
for any lumped circuit the algebric sum of current entering or leaving a node is equal to zero at any instant of time.

It depends on conservation of charge.



$Q_1 + Q_2 = Q_3 + Q_4$ $\quad I_1 + I_2 = I_3 + I_4$

Example ①

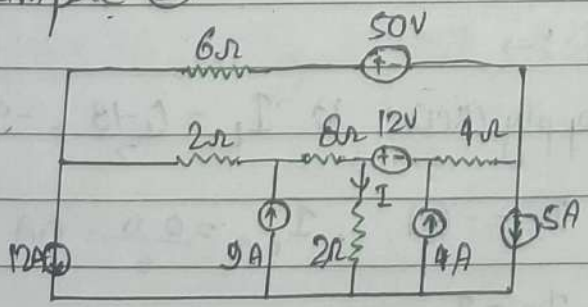


Solve.

$8 = 12 + I$

$I = 12 - 8 = 8 - 12 = -4\,A$

Example ②

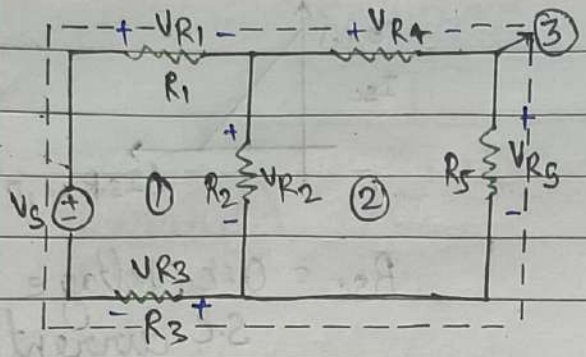


total incoming current = tot. out current

$9 + 4 = 12 + I + 5$

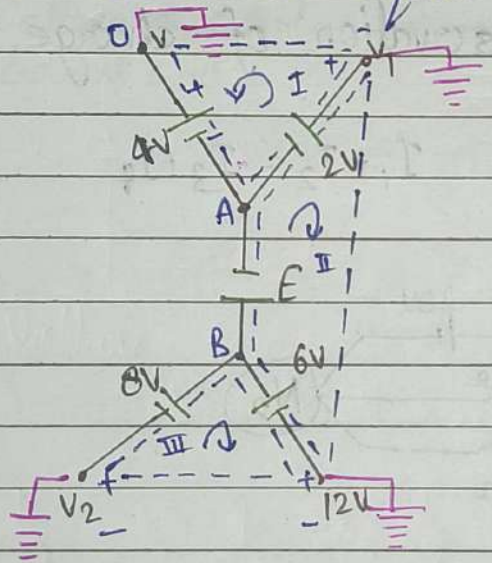
$I = 13 - 17$

$I = -4\,A$



→ The sum of voltage drop across in each element in any loop is equal to zero. it is based on conservation of law.

The algebric sum of node voltage for any closed nodes is equal to zero.



$2 - V_1 - 0 - 4 = 0$

$[V_1 = -2V]$

$V_1 + 12 + 6 - E + 2 = 0$

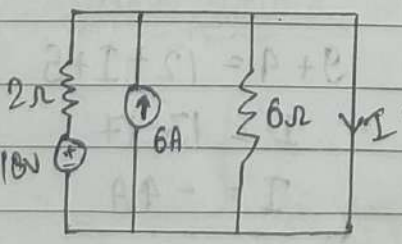
$-(-2) + 12 + 6 - E + 2 = 0$

$[E = 22V]$

$8 - 6 - 12 + V_2 = 0$

$V_2 = 10$

Ques Find the value of I :→



apply (KCL) → @ $I_L = \frac{0-18}{2} = -9A$

$I_S = -6A$, $I_{6\Omega} = \frac{0-0}{6} = 0A$

$I_R = I$

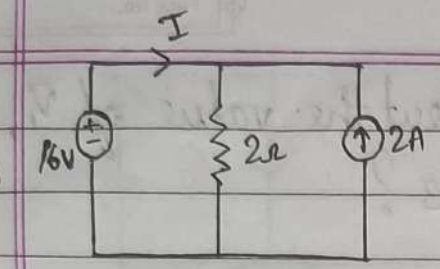
$-9 - 6 + 0 + I = 0 \Rightarrow -15 + I = 0$

$[I = 15A]$

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b)



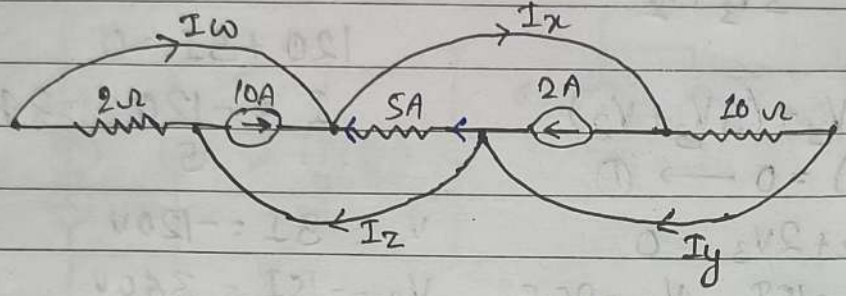
$I_{2\Omega} = \frac{16}{2} = 8A$

current entering = current leaving

$I + 2A = I_{2\Omega} = 8A$

$I = 8 - 2 = 6A$

c)



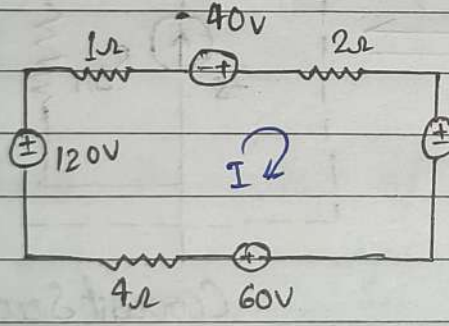
$10 + I_\omega + I_{5\Omega} = I_x$ | $I_z = I_\omega + 10$

$I_\omega = 0; I_x = 10A$ | $2 + I_y - I_5 - I_z = 0$

$I_z = 10;$

$I_z = I_y + 2 \Rightarrow \boxed{I_y = 10 - 2 = 8A}$

d)



$-I(1) + 40 - I(2) - 20 - 60 - I(4) + 120 = 0$

$(-1-2-4)I + (40+20-60+120) = 0$

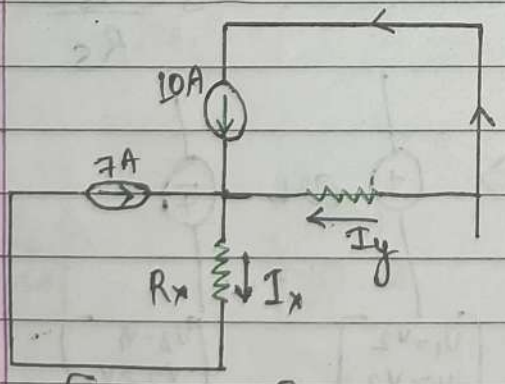
$-7I + 80 = 0$

$7I = 80$

$I = \frac{80}{7} \simeq 11.43A$

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e)



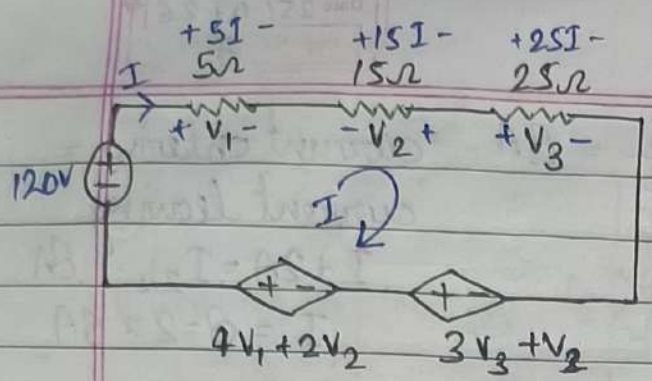
find the value of I_x and I_y.

solve → $\boxed{I_x = 7A}$, $I_y = -10A \Rightarrow 7 + 10 + I_y = I_x$

$7 + 10 + I_y = 7$

$\boxed{I_y = -10A}$

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find out the value of V_1, V_2, V_3?

Using KVL

$120 - V_1 + V_2 - V_3 + (3V_3 + V_2) + (4V_1 + 2V_2) = 0 \longrightarrow (1)$

$\Rightarrow 120 + 3V_1 + 4V_2 + 2V_3 = 0$

$V_1 = 5I,\ V_2 = -15I,\ V_3 = 25I$

$120 + 15I - 60I + 50I = 0$

$120 + 5I = 0$

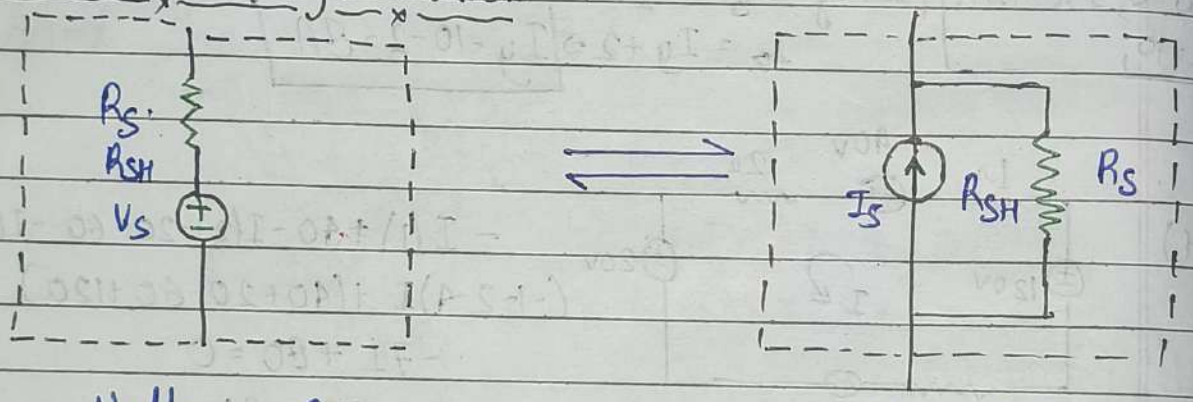
$I = \frac{-120}{5} = -24A$

$V_1 = 5I = -120V$

$V_2 = -15I = 360V$

$V_3 = 25I = -600V$

Source Transformation →



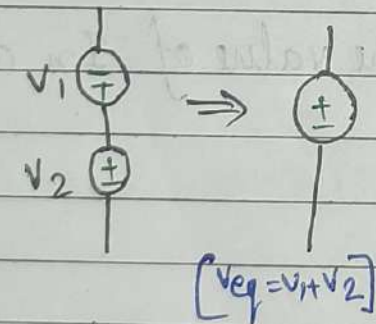
Voltage Source

$V_s = I_s \cdot R_{SH}$

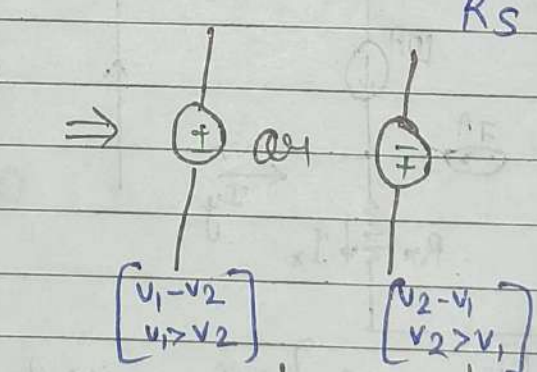
Current Source

$I_s = \frac{V_s}{R_s}$

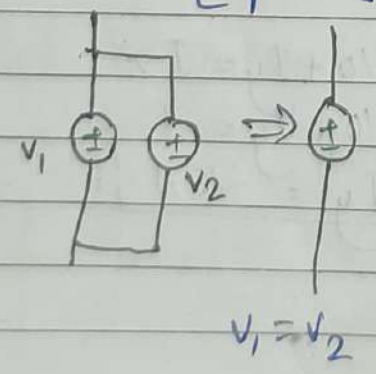
a)



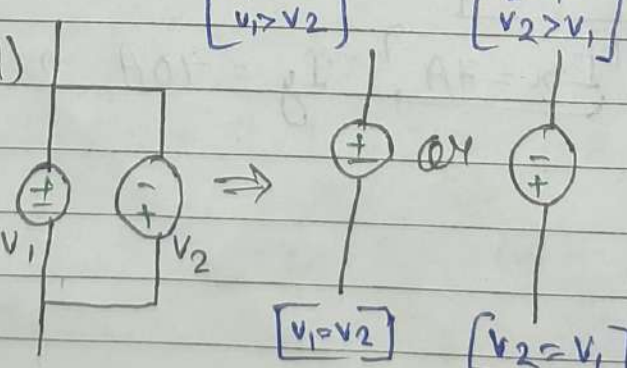
b)

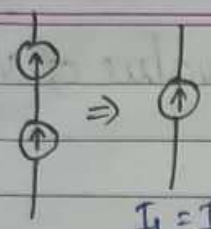


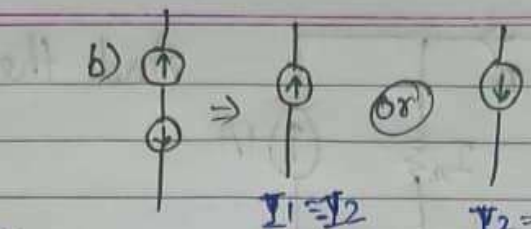
c)

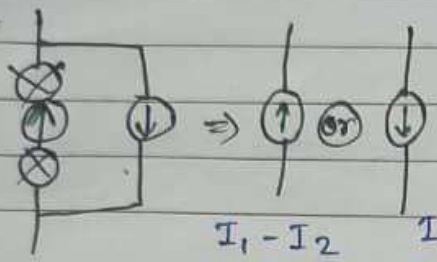


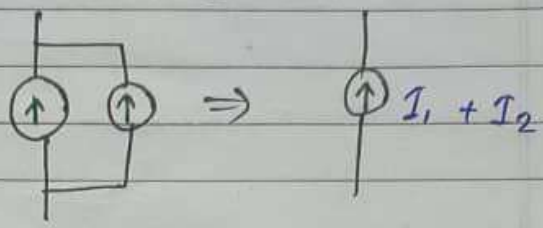
d)

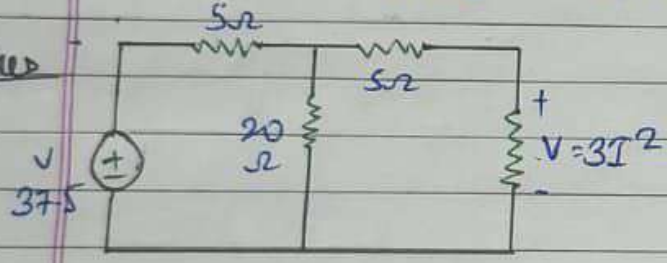


a)  $I_1 = I_2$

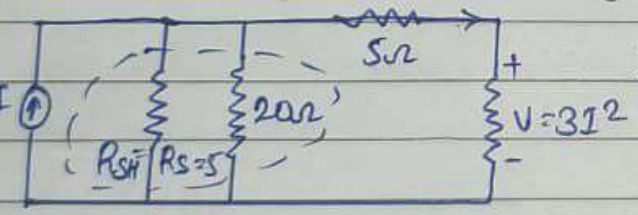
b)  $I_1 = I_2$ $I_2 = -I_1$

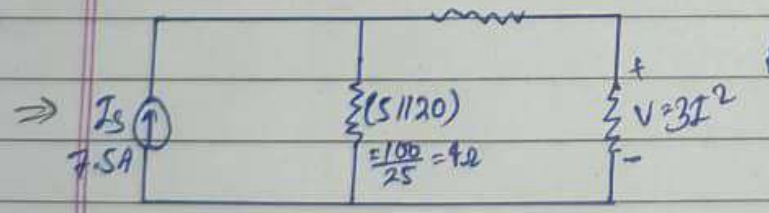
c)  $I_1 - I_2$, $I_1 > I_2$ $I_2 - I_1$, $I_2 > I_1$

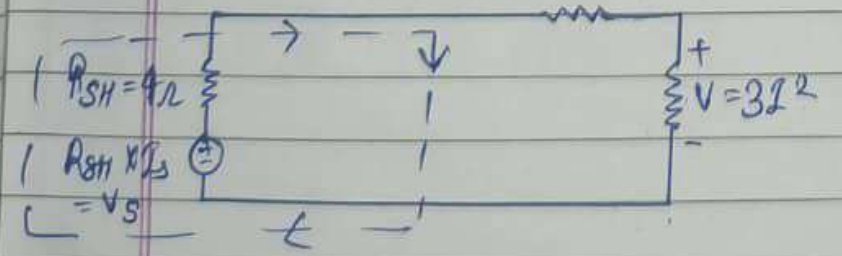
d) 

Ques 

find the value of V using source transformation?

Solve→  $I = \frac{V_s}{R_s} = \frac{37.5}{5} = 7.5A$

 $R_{SH} = I \times R_S = 4I$



$30 - 4I - 5I - 3I^2 = 0$

$3I^2 + 9I - 30 = 0$

$I^2 + 3I - 10 = 0$

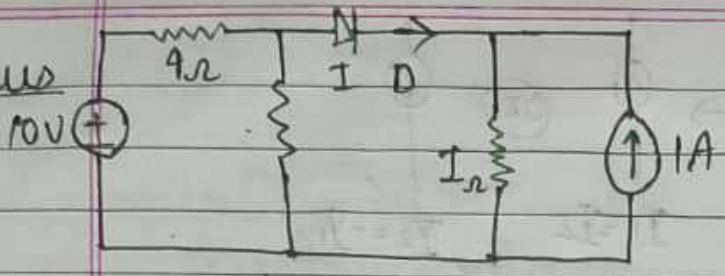
$I^2 + (5-2)I - 10 = 0$

$I^2 + 5I - 2I - 10 = 0$

$I(I+5) - 2(I+5) = 0$

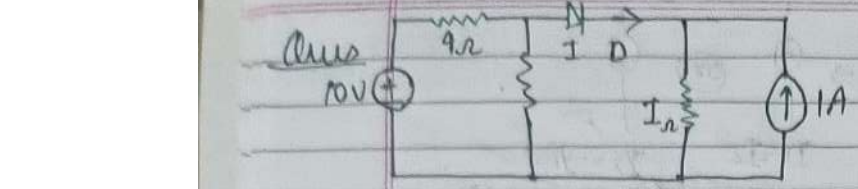
$I = -5 \times \quad I = 2, \boxed{V = 3I^2 = 12}$

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Ques 

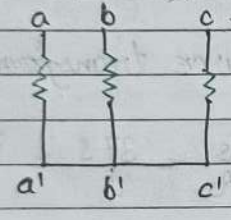
find the value of v?

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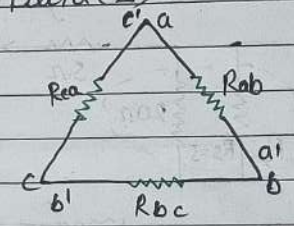
Ques 

find the value of v?

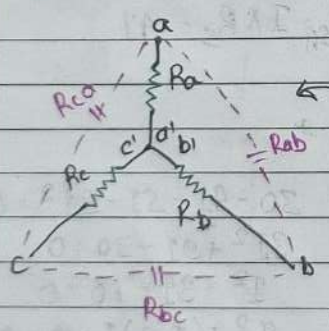
Star and Delta Connection →



b) Delta (Δ) →



a) Star (Y or Y) →



star to delta transformation →

$$\left[R_{ab} = R_a + R_b + \frac{R_a \times R_b}{R_c}\right]$$

$$\left[R_{bc} = R_b + R_c + \frac{R_b \times R_c}{R_a}\right]$$

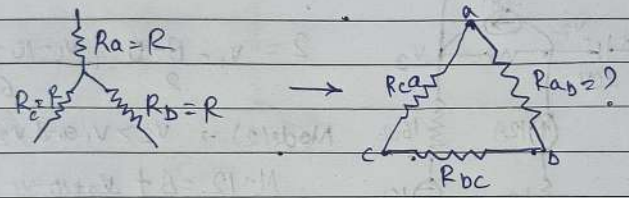
$$\left[R_{ca} = R_c + R_a + \frac{R_c \times R_a}{R_b}\right]$$

$$R_a = \frac{R_{ab} \times R_{ca}}{\Sigma R} \qquad \Sigma R = R_{ab} + R_{bc} + R_{ca}$$

$$R_b = \frac{R_{ab} \times R_{bc}}{\Sigma R} \quad , \quad R_c = \frac{R_{ca} \times R_{bc}}{\Sigma R}$$

Ques Build the connection if the connection is balance then find star to delta and delta to star connection?

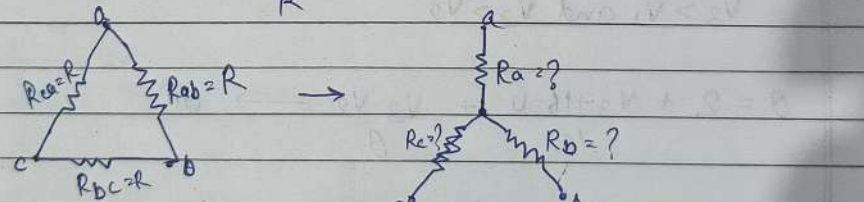
i)



$$R_{ab} = R_{ba} = R_{ca} \; ; \quad R_{ab} = R_a + R_b + \frac{R_a \times R_b}{R_c}$$

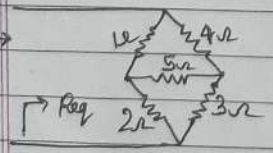
$$R_{ab} = R + R + \frac{R^2}{R} \Rightarrow \boxed{3R \Rightarrow R_{ba} \Rightarrow R_{ca}}$$

ii)



$$R_a = \frac{R_{ab} \times R_{ca}}{\Sigma R} = \frac{R \times R}{3R} \Rightarrow \boxed{\frac{R}{3} \Rightarrow R_b \Rightarrow R_c}$$

Ques →



⇒

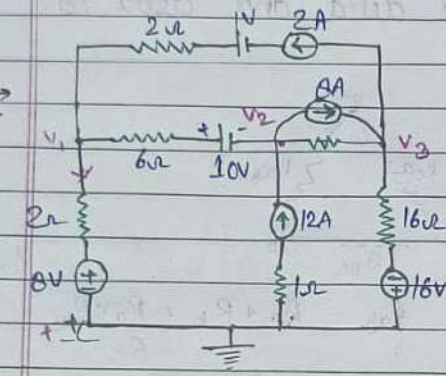
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④ Nodal Analysis :→

→ It is based on kirchoff current law and Ohm's law.

→ It is applicable on both plane and non-planner

Ques→



Node(1) →

$V_1 > V_2$ and $V_3 > V_3$

$$2 = \frac{V_1 - 8 - 0}{2} + \frac{V_1 - 10 - V_2}{6} \rightarrow$$

Node(2) → $V_2 > V_1$ and $V_2 > V_3$

$$N \rightarrow 12 = 8 + \frac{V_2 + 10 - V_1}{6} + \frac{V_2 - V_3}{8}$$

→ (II)

Node(3) →

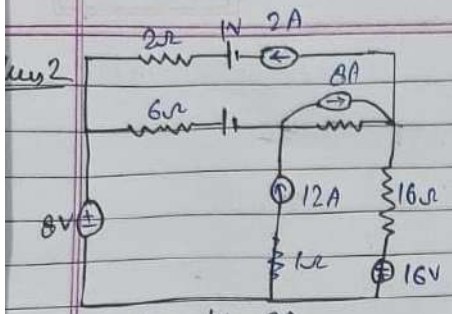
$V_3 > V_1$ and $V_3 > V_2$

$$8 = 2 + \frac{V_3 + 16 - 0}{16} + \frac{V_3 - V_2}{8} \rightarrow \text{(III)}$$

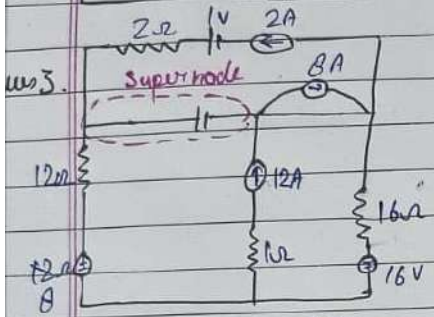
$V_1 = 8V$

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Ques 2



Ques 3.



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Ques 4.

